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Nicolette Lakemond and Gunnar Holmberg

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Nicolette Lakemond
Department of Management and Engineering
Linköping University
Linköping, Sweden
nicolette.lakemond@liu.se

Gunnar Holmberg
Department of Management and Engineering
Linköping University
Linköping, Sweden
gunnar.holmberg@liu.se

***Abstract*— In contrast to retrospective approaches, as often used in management research, this paper outlines an approach to create knowledge on partly unknown futures contributing to the shaping of emerging phenomena while at the same time creating new knowledge. Grounded in the logic of prescience, the approach is illustrated by a case study representing a future complex intelligent system in the area of public safety. The findings show avenues to arrive at evolving conjectures regarding the management of complex intelligent systems. The derived approach is iterative and abductive with a representation of the phenomenon and its related novelty at the heart of the inquiry, and evolving conjectures as input as well as output of iterative and abductive cycles. It relies on collective acts of insight through a dialectical process of interaction between stakeholders to reach cumulative synthesis. The approach could bring value to management research, representing rigorous and relevant ways to bring societal value and help shape a better future when AI is becoming increasingly important in society.**

Keywords—artificial intelligence; complex intelligent systems; prescient approach; engineering management; representations; conjectures; collectivity

I. INTRODUCTION

Capturing an emerging phenomenon and contributing with theories shaping a partly unknown future is a pressing and challenging task. Today, many academics believe that the research community needs to suppress a strong fetish for theorizing for its own sake and favour research that addresses important phenomena and is seeking to impact on practice [1]. However, many of such important phenomena, such as related to evolvements in the field of artificial intelligence (AI) are still emerging and reside in a partly unknown future that is shaped in the present but also by the legacy of the past. Therefore, an important quality for understanding an emergent phenomenon is to connect the past, present and future. Shaping a future with AI entails probing into the future, i.e. following a prescient approach that enables *"anticipating and influencing the type of managerial knowledge needed to deal with coming societal and organizational concerns"* [2, p. 13]. Generative reasoning, abduction, and disciplined imagination have been associated with such creation of an understanding in the partly unknown [3], building on relatively fluid, open and mutual learning environments for learning, shared reflection, and co-creation between a variety of stakeholders [4]. Yet, in contrast to the engineering disciplines, in management research, approaches to create knowledge on a (partly) unknown and novel phenomenon are not very well established, and more insights into how such approaches can be designed and employed appears necessary.

Following this, and with the overall aim to spur a debate on how the future of management with AI as part of complex and increasingly intelligent systems can be shaped, the purpose of this paper is to explore underpinnings of a prescient research approach allowing for structured managerial knowledge creation on new phenomena in the partly unknown cf. [5].The paper focuses specifically on approaches for emerging intelligent technologies and autonomy of systems that increasingly rely on AI solutions. Although insights have started to appear, and the research community is mobilizing its interest, most of the early contributions still focus on conceptual arguments and development (see e.g. [6], [7]) rather than insights that build on structured empirical observations of an emerging phenomenon and include forward-looking knowledge creation. Deeper insights are sometimes offered for relatively stand-alone applications of AI, while long-term benefits of AI are expected in complex settings with essentially more intriguing implications for management and requiring a broader understanding [8][9].

Currently, the emergence of intelligent technologies, relying on AI, is on top of the agenda in a large variety of industries. These include the automotive and aviation industry with autonomous vehicles, but also other sectors such as communication, finance and life sciences. Many intelligent technologies are implemented in systems that fulfil crucial functions in society. In society and industry, there is a sense of urgency to address technological and industrial capacity related to the uptake of AI, the socio-economic consequences brought about by AI, and ethical and legal frameworks (e.g. [10], [11]). Scholars have started to realize that it is necessary to develop new management approaches that can address the complex and multidimensional aspects and help shaping the desirable outcomes of emerging technologies [12]. This is particularly important as the emergence of intelligent technologies are not only a product of the phase in which these technologies are in operation, but largely influenced by the visions of those who design technologies [13]. This requires new research approaches that allow for generative knowledge creation based on studies of a phenomenon partly before its actual implementation [9] [12].

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Based on a case study of a research arena aiming at creating an understanding of future complex and increasingly intelligent systems, this paper formulates such a prescient approach that addresses a partly unknown phenomenon and anticipates on the management implications and consequences of AI in complex intelligent systems (CoIS).

II. The Emergence of Complex Intelligent Systems

A. Shaping a Future with AI

As part of the transformation towards the implementation of intelligent technologies in complex systems, European as well as US policy documents have acknowledged the need to actively manage risks and challenges to be able to draw benefits from advancements in AI. In a report by the EU Commission from 2018 it is stressed that *"the way we approach AI will define the world we live in".* In the US, a similar awareness is reflected in a report from the Executive Office of the President, National Science and Technology Council Committee on Technology, from October 2016: *"In the coming years, AI will continue to contribute to economic growth and will be a valuable tool for improving the world, as long as industry, civil society, and government work together to develop the positive aspects of the technology, manage its risks and challenges, and ensure that everyone has the opportunity to help in building an AI-enhanced society and to participate in its benefits."* Similar calls were made during the Spring 2023 when Open AI expressed concerns regarding the difficulties to navigate the development of general AI in the unknown to increase the chances to benefit from AI while suppressing the threats. They expressed among others *"We currently believe the best way to successfully navigate AI deployment challenges is with a tight feedback loop of rapid learning and careful iteration".* The policy institute Future of Life, perhaps most known for its "Asilomar AI principles" has, based on what OpenAI has communicated, published an open letter that calls for thoughtfulness and new ways to ensure that AI will make societal infrastructures more safe, robust and reliable.

B. AI implementation in Complex Systems

This paper looks beyond AI as a new technology, and builds on the assumption that AI needs to be considered in a wider context than the AI application itself. Although expectations on AI are high and many aspects, including societal and ethical aspects are discussed related to either AI technology or fully autonomous systems [10], [11], the vast majority of available AI applications are still standalone [15] and not an integrated part of systems. However, it evident that AI is becoming a part of complex systems, such as energy, healthcare, and other infrastructures critical to society and building on many different technologies, components and subsystems [16] [8]. In addition, it appears that Generative AI increasingly is seen as a potential contributor to complex intelligent systems. This results in systems that increasingly display intelligent behaviour, i.e. systems are capable of analyzing their dynamic context and take actions, and have some degree of, but not full, autonomy to achieve specific goals. Despite this, the debate on AI is biased towards potential hazards with the future existence of fully autonomous systems [17] rather than the effort necessary for shaping systems in which human intelligence, organizations, and system autonomy are co-existing. Such systems are characterized by collaborative intelligence [18] rather than full autonomy and would direct the attention towards understanding the role of humans, selected system boundaries and the bounds of AI solutions. This also applies to envisioned future fully autonomous systems. As an example, an autonomous vehicle will need to be understood in the larger context of the traffic system, other vehicles, pedestrians, surrounding etcetera. This provides a strong case for the need to develop an understanding of the management implications related to the emergence of a wider complex intelligent system [8]. Such an understanding reflects a view on machines, artifacts and human beings that exist in constantly shifting relationships that are what has been referred to as "mangled" together in unforeseeable ways [19]. It can thus be argued that, in the context of the emergence of CoIS, particularly the evolution of fluid boundaries between humans and (technical) systems, their interaction and the interplay between human authority and systems autonomy are central to understand management implications [8].

C. New Management Challenges for CoIS

Understanding the implementation of intelligent technologies requires a deep insight into the phenomenon of complex systems, the situatedness of these systems [18] as well as the new prerequisites provided by emerging technologies [14]. Complex systems have received increasing attention due to their specific innovation characteristics, but also as they form the backbone of our societal critical infrastructure [20], [21]. For instance, current systems such as healthcare, transportation, security, and public safety are complex systems based on large numbers of interacting elements, many attributes of interest and many stakeholders involved with different objectives and needs [22]. Stakeholders may be acting and reacting and result in system responses that are difficult to predict. In addition, the system may be subject to phenomena that are emergent, i.e. the system itself may be self-innovating [6]. With the introduction of intelligent technologies, AI can potentially perform a range of interdependent tasks integrated in the system, perhaps outside the direct control of system designers and system integrator firms [7]. Consequently, the challenges related to complex system design and operation may alter drastically and new competences, organizational forms and approaches, and system architectures will be required [23]. The introduction of AI creates conditions related to learning, autonomy, and adaptability aspects of complex systems [8]. It alters existing engineering design practices where approaches relying on functional requirement breakdown and the search for corresponding solutions [24] have become insufficient. The focus on system design rather is changing into an endeavour centred around creating a trainable data-driven system that could respond to evolving needs [8].

III. A Prescient Approach for Shaping New Engineering Management Practices

A. Prescience as a forward-looking approach

With current developments and challenges in society, the need for creating knowledge on emerging and partly unknown phenomena has never been greater [2], [3]. While engineering disciplines are well-equipped with approaches to probe into the future, much of the current approaches in management research are not very well suited to do so. Scientific studies often build on retrospective or perhaps *insitu* studies, focusing on theorizing on past practices that are formulated to inform and inspire a larger audience of organizations on 'best practices'. While such studies may provide interesting insights

into current practice, it is unclear if the past and the present can provide reasonable guides to the future [25] to identify and address anomalies that are at least partly unknown and difficult to foresee. These anomalies are novel and emerging phenomena that cannot be readily explained by existing knowledge and thus require generative learning in a cyclical and abductive process of identifying, generating, and evaluating aspects of the phenomenon [3].

A more prescient, forward-looking, focus is present in several established research methods that particularly address the future. For example, studies the field of technology forecasting or foresight place a particular emphasis on the future. Common methods include scenario planning and strategic analysis [26]. This has for instance resulted in extensive approaches that may help analyze future-oriented technology and its consequences. Some particularly address innovation pathways, such as related to nanotechnology [27]. In addition, a relatively large body of literature on sociotechnical systems tends to focus on futures that are accessible to or constrained by a certain regime and may highlight important implications on the relationship of the past and future as well as technology and organizations [28]. In other research streams the unknown has received attention. For instance, in design theory [29] the unknown in the future of technological design has a central focus, and design approaches building on structuring the unknown have been proven to facilitate probing generatively into new phenomena [5], [30].

Although these contributions show that it may be possible to generate knowledge in future and partly unknown domains, they require a prescient approach relying on solid pillars to anticipate and influence managerial knowledge needed to deal with upcoming societal and organizational concerns [2]. This includes an orientation toward prospection for making informed projections and inferring what aspects need attention, most likely supported by disciplined imagination and the generation of new insights through representations, abductive hunches or conjectures, collective approaches, and dialectical processes [3], [31]. This is reflected in Table I.

B. Finding representations of the phenomenon

Much of existing management research captures knowledge based on studies of events that have already taken place, e.g. through interviews with key informants or more large-scale quantitative surveys. A study of a partly unknown phenomenon calls for ways to capture something that cannot be readily observed. Probing into such a phenomenon involves disciplined imagination through trial-and-error thinking and deliberately introducing diversity into problem statements [31]. Weick [31] suggests a close parallel to the process of theorizing and evolutionary processes, i.e. guided by representations of the context, not the context itself, and using these representations to make inferences on the actual phenomenon. When such representations can be studied, they can serve as a foundation to make projections of the future and thus inferences on what needs to be addressed in theory generation. As Corley and Gioia [2, p. 25] put it: "Prescient scholars would, therefore, develop an orientation toward prospection—they would assume the role of making informed projections about coming issues, act as if those issues have manifested, and then infer what theoretical domains need attention or invention." Such prospection needs to be designed with close attention to the context of creating societal value [32], while this context is not yet completely present and cannot be easily prospected. This thus involves sense-making as well as sense-giving [2] in relation to the phenomenon as well as its context. This includes initially divergent observations that describe and confirm the anomaly as well as the creation of convergent views of the phenomenon, its context and characteristics that provide confirmation that the phenomenon exists [3] However, noticing and confirming a phenomenon and its context requires empirical observations of something that does not exist fully and/or is implemented fully yet. Finding suitable representations and gathering evidence on the nature, context and what is known about the phenomenon thus becomes a central aspect in a research approach that can shape future management approaches.

C. Abduction and conjectures

Shaping future engineering management approaches is not a one-time endeavour but requires consistent attention over a duration of time. Neither inductive nor deductive approaches seem to be suitable as they are typically not generative in nature [3]. Rather, an abductive approach provides opportunities for knowledge creation as an iterative long-term endeavour. Abduction allows moving from the underexplained towards plausible explanations through a process of systemic combining [3], [33], [34]. It allows for a generative process in which original insights are modified following for instance new and unanticipated findings but also new insights that are gained throughout a knowledge creation process. Such a research process is path-dependent and nonlinear, based on combining efforts of matching theory and reality [34]. An abductive process can capture novel phenomena as alternating stages of divergence and convergence including the generation of conjectures that provide plausible, but not yet valid, explanations of the phenomenon [3]. These conjectures rely on disciplined imagination where *"…the discipline in theorizing comes from consistent application of selection criteria to trial-and-error thinking and the imagination in theorizing comes from deliberate diversity introduced into the problem statements, thought trials, and selection criteria that comprise that thinking."* [31, p. 516]. It has been argued that better theories are generated when they build on a great number of diverse conjectures requiring heterogeneity among thought trials [31].

D. Collectivity and dialectics

A prescient approach needs to pay close attention to the context of creating societal value [32]. Close interaction between a diverse range of actors, including reflective practitioners, researchers from different disciplines, policymakers, and decision-makers, in an engaged scholarship mindset [35] is expectedly imperative to capture such a context and selecting and evaluating evolving conjectures from a variety of perspectives. In this context, it has been referred to the importance of collective acts of insight through a process of cumulative synthesis in knowledge generation on novel phenomena [3].

Probing into the partly unknown future benefits from dialectical processes that enable scenarios and variations to address potential incompatibilities and dilemmas which represent the difficulties in moving forward into partly unknown directions cf. [36]. These incompatibilities and dilemmas can emerge as tensions and are typical for complex systems that become increasingly intelligent [9] and operate in a partly unknown future context. These emanate from evolving societies with for instance many stakeholders and

important constraints and potentialities ingrained in technology and systems architecture [23].

Dialectics originates from Hegelian perspectives and describes collective processes were opposites or conflicting perspectives form a tension that can be resolved towards a new state, i.e. a synthesis. It has been acknowledged that such a process may not always result in a synthesis but also for instance emerge as power struggles and conflict or as an evolvement of events without a clear synthesis [37], [38]. Dialectical processes are thus dynamic and iterative and emerging events often meet new tensions in the social context of organizations [39]. Tensions are thus shaping, but also being shaped by, organizational achievements and emerging situations in a process that is evolving in cycles [40], [41]. Tensions could be used in an intentional way to influence and form the future, resembling a change process that relies on teleological as well as dialectical elements [42].

Abductive approaches to develop conjectures have been strongly associated with collective dialectical processes. It has been argued that *"However, the value of collective abduction is not just one of adding another level but lies in the fact that collective abduction entails a dialectical process of interaction that is different from an individual-level evolutionary process"* [3, p. 690]. The collective dimension is also stressed in the "engaged scholarship" approach [35], although perhaps more implicitly, by expressing that new knowledge must be created in the context of its application and through the interaction of academic researchers and informants or reflective practitioners that are deeply ingrained in this context [32].

TABLE I. THE PRESCIENT APPROACH

Elements of a prescient approach	Explanation
Finding respresentations of the phenomenon	Representation of the phenomenon were observations can be made that form the basis for making inferences on the actual phenomemon. Serve as a foundation of projections of the future that guide the shaping of this future
Abduction and conjectures	A generative process in which insights are continuously modified following new and unanticipated findings. Based on the generation of evolving conjectures as plausible, but not yet valid, explanations of the phenomenon. Disciplined imagination through consistent application of selection criteria to trial-and-error thinking and deliberate diversity introduced into problem statements and thought trials.
Collectivity and dialectics	Collective acts of insight through a dialectical process of interaction to reach cumulative synthesis

IV. METHODS

The empirical evidence presented in this paper is based on a single case cf. [43] of a system that represents a future intelligent complex system. This specific case, WARA-PS, focuses on public safety and provides a realistic, large scale and industrially relevant demonstration environment as part of the development of long-term future technology capabilities in a system of systems context. WARA-PS uses scenarios that focus on keeping society and its citizens safe, such as search and rescue missions when people are in emergency situations at sea. It involves multiple technologies and functionalities with varying autonomy to be integrated into a system of systems in which autonomous vehicles (e.g. submarine vessels, surface vessels and drones) collaborate together for achieving search and rescue missions. WARA-PS has evolved from demonstrating collaborative missions, via dealing with unknown situations to currently focus on resilience at mission level. The demonstrations are selected with the intention and understanding that other emerging CoIS would benefit from gained insights and knowledge in which e.g. resilience is expected to rely on multiple technologies and functionalities collaborating with humans.

The study is based on 11 interviews (around 9.5 hours) with key participants with expert knowledge in development of complex and increasingly intelligent systems, especially with applications and or constituent systems in the domain of search and rescue. The interviewees have roles in academic research, industrial R&D, and public sector. Additional complementary secondary data was gathered as well as (participant) observation performed (more than 400 hours) by one of the co-authors in planning and demonstrations of the system. The case illustrates many of the future characteristics of future CoIS. Although in an experimental setting, it can be used, as we intend to show, as a representation of the phenomenon, serving to capture the many variables and intricacies involved. The single case allows us to go deeper into one case rather than exploring many superficially cf. [34]. The data were triangulated among the different sources with the purpose to increase the validity of our findings cf. [44]. The analysis has been performed in recurrent interactions between the researchers involved, recurrent contacts with key informants deeply involved in the case as well as a systematic engagement with the data emanating in an empirical narrative. The focus has been on formulating a prescient approach, i.e. showing how a representation of an emerging phenomenon can serve to create knowledge on in an emerging but not yet fully realized context. In the next section, the case, its context and relevance is described in more detail.

V. FINDINGS

A. The Case of WARA-PS as a respresentation of the phenomenon

Many countries have initiated major research initiatives on AI to create sound technology foundations. Initiatives are often accompanied by specific national strategies that are describing the issues that need to be addressed. For instance, in Sweden, the Swedish government released its national AI strategy in May 2018, reflecting a national approach to AI and creating a basis for further policy actions and priorities. Sweden is only one of many countries pursuing the potential benefits with AI but places well on indexes that compare different countries. For instance, in the third edition of the Government AI Readiness Index published in 2023 by Oxford Insights and the International Research Development Centre (IDRC), Sweden is ranked number 13 out of 181 countries. An overview of the ongoing AI landscape in a global perspective, such as provided by AI watch of the European Commission shows a dominance of the USA and China in terms of AI players and activities. This may not be surprising as both are very large countries.

In Sweden, a major research initiative is ongoing in the Wallenberg AI, Autonomous Systems and Software Program (WASP), in which technology and software development for future intelligent systems is addressed. The program has a

research budget of 6.1 billion SEK (approximately 0.55 billion Euro) and is the largest in Sweden ever. The program has since 2020 been complemented by a separate program called WASP-HS with a specific focus on Humanities and Society and a mission to study the impact of the technology transition resulting from the development of AI and autonomous systems in terms of ethical, economic, labour market, socio-economic and legal aspects.

As part of the WASP program, several WASP Research Arenas (WARAs) are created. The research arenas have the purpose of focusing on industrially relevant scenarios covering one or several defined research challenges. WARA Public Safety (PS) was the first of these arenas specifically aiming to provide a relatively realistic, large scale and industrially relevant demonstration environment using scenarios. It represents a complex system of systems in which a diversity of industrial and academic actors with different competences are involved, and in which human-based and artificial intelligent technologies are collaborating in a system of systems context. The arena brings together autonomous systems (unmanned arial vehicles (UAVs), and unmanned surface vehicles (USVs)), decision frameworks to execute search and rescue missions, and human operations. It involves areas such as motion planning, situation awareness, control theory, cloud computing, positioning and navigation, image processing, and resource management.

The purpose of the demonstrations is to stimulate the creation of knowledge on how the different technologies and approaches can be combined and the creation of new system building approaches that meet increasing demands on a system's ability to respond to dynamically changing needs, support unforeseen situations. An important goal is to understand how humans and intelligent systems can solve incompletely defined missions together in an effective and efficient way.

B. Abduction and two conjectures in WARA-PS

Based on the characteristics of WARA-PS, through interactions with key informants, and previous insights into complex and increasingly intelligent systems such as developed in previous studies on for instance avionics [9], two preliminary hunches about the anomalies of the phenomenon were identified. They can be considered as initial conjectures. These include (1) combined human authority and systems autonomy, and (2) combining safety criticality and generativity. These conjectures formed an initial starting point and were used to derive engineering management insights and implications, including new and refined conjectures. For instance, related to the problem, phenomenon, and context, the hunches related to the first conjecture imply among others that new approaches for supporting the collaboration between humans and intelligent system need to be addressed, and that the boundaries between humans and systems tend to become more fluid. In addition, humans and systems are innovating together during operation, i.e. they engage actively in a design process. Early observations in WARA-PS showed that such collaboration relies on long-term preparations where system architectures and DevOps are central. It also involved purpose-driven considerations, i.e. taking into account a higher purpose and expanding the complexity of decisions to be made.

Related to the second conjecture on the combination of safety-criticality and generativity, the combination of the safety-critical operations together with a generative ecosystem with fluid boundaries were found to be important. Here it appeared that development took place in planned and emergent ways, with different time horizons that were supported by planning and reactive development approaches. The focus on safety and security were being complemented by purposeful aspects.

The initial conjectures and insights from the observations through the study provide guidance in the further exploration toward updated conjectures in a process of matching, direction and redirection. For instance, in relation to conjecture 1, it was found that the combination of humans and systems create a complex interface that does not lend itself to complete modelling as the engineering approach. Rather, the combination of data-driven and model-based methods appeared as a promising route forward. WARA-PS responded to this by increasing its focus on data generation as an important stimulus for continued development. This also led us to consider an expansion on the early conjecture that innovation is data-driven, to include generative data and 'innovation-driven data generation', reflecting that the use of machine learning algorithms often is starving from lack of useful and relevant data. It was also found that the fundamentals of future human-system interaction was evolving and rather vague when it came to complex and incompletely defined situations. This provided insights on a necessity to find novel methods to manage ecosystems and combining the use of data and models in a new context where new actors such as data factories and algorithm makers join the ecosystem.

C. Collectivity and dialectics in WARA-PS

In yearly demonstrations of evolving scenarios from 2018 and onwards new insights into the workings of the collaborative system-of-systems was created while a large number of stakeholders joined the collective endeavour. During 2023, the demonstration involved a whole week focusing on search and rescue missions at sea, finding missing people using virtual platforms, and delivering medical supplies with the help of UAVs. During the demonstrations, a wide variety of stakeholders were invited to discuss research, demonstrate new results, collect data and not the least inspire each other.

As one of the emerging insights through the collective involvement and commitment in WARA-PS, it was found that the fundamentals of future human-system interaction was evolving and rather vague when it came to complex and incompletely defined situations. This provided insights on a necessity to find novel methods to manage the community representing the ecosystem and combining the use of data and models in a new context where new actors such as data factories and algorithm makers join the ecosystem. Further, linked to collectivity, dialectic positions evolve within the community, sometimes polarizing into those focusing on the possibilities and those on the difficulties. It appeared that the focus on the unknown as well as on achieving resilience offered a basis for resolving such dialectic tensions. Still, it seemed challenging to get the full community joining in on such a discussion.

The evolving insights into the two conjectures are summarized in Table II.

TABLE II. CONJECTURES AND EMERGING INSIGHTS

Conjecture	Combined human authority and systems autonomy	Combining safety criticality and generativity
Initial insights on the phenomenon in its context	Collaboration between humans and intelligent systems Fluid boundaries between humans and systems Humans and systems designing together during operation	Combination of safety critical operations with generative ecosystem of actors Innovation: - planned and emergent - several time horizons - planned and reactive development approaches - data driven Safety and security are being complemented by purposeful aspects
Emerging insights leading to updated conjectures and implications	Human-system interface cannot be completely modelled Combination of data-driven and model-based methods are promising route forward Long term approaches (Architectures, DevOps etcetera) enable innovation in operation Purposeful considerations expand complexity of decisions in development and operation Fundamentals of future human-system interaction is evolving and vague when it comes to complex incompletely defined situations	Combination of safety criticality and generativity enforces combination of strict methods and open generative methods Generative data and methods as important contributors to safety criticality Innovation-driven data generation Hybrid engineering management models as enablers to support recombination and use of available resources between criticality levels Ecosystem of actors with fluid boundaries

D. Insights for the prescient approach

The prescient approach aims at deriving an understanding of a phenomenon, including potential or observed relationships as part of an ongoing knowledge creation and design endeavour. In WARA-PS, this could for instance concern the insight into the evolving and indeterminate human-system interaction in incompletely defined situations, or the notion of innovation-driven data generation contrasting and complementing a common belief of data-driven innovation in the context of AI.

Although WARA-PS was designed for the purpose of creating insights on future search and rescue benefitting from intelligent technologies, it lends itself to meaningful prescient studies on the future of management for such systems as well. The evolutionary nature of WARA-PS resonates with the abductive approach where insights from both WARA-PS and our studies contribute to shape the next steps of WARA-PS. The understanding of different representations, phenomenon aspects and conjectures have evolved dynamically. The context in which hunches and conjectures have evolved is rich with opportunities to discuss and elaborate on insights strengthening the ability to identify and detailing plausible conjectures. A prescient approach includes discussions on uncertainties in observations, insights on impact, and on the overall phenomenon and its context, as well as conditions for applicability.

As a limitation of the prescient approach, the insights will always be conditioned by the representations and the envisioning of the future context and the phenomenon. The overall rigor of the prescient approach could thus be strengthened not only by identifying a variety of representations, but also by complementary retrospective and currently observable contexts, for instance in different domains. Such a variety of sources for insights, could provide continuously inform and evolve conjectures into more precise frameworks and eventually theories. Table III outlines the prescient approach.

TABLE III. APPLYING THE PRESCIENT APPROACH

Prescient approach	Using the approach
Finding respresentations of the phenomenon	Focus on expected future properties Consider the phenomenon and its context Consider the extent to which the future is made observable Make sure that known and unknown aspects can be observed while emerging
Abduction and evolving conjectures	Create the possibility to tap into a variety of sources that are observable, envisioned complemented with retrospective perspectives to capture legacy Consider expanding representations for the least known aspects of the phenomenon
Collectivity and dialectics	Be immersed in the phenomenon and context Create interactions with actors Participate in observable events like the demonstrations and their planning Tap into the visions of the involved actors Create discussion on the anticipated uncertainties in insights and how this impacts the evolving conjectures, e.g. under which conditions the insights apply

VI. CONCLUSIONS

In this paper a prescient, forward-looking, approach is suggested and explored to assist in knowledge creation and shaping of a future with AI in complex intelligent systems. As such, the approach is reflecting a design approach, involving a new model of collective action [45].

This prescient approach aims at facilitating the exploration of novel phenomena in the partly unknown cf. [2]. It is an iterative, abductive and collective approach with at the heart of knowledge creation the identification of representations of the phenomenon. Finding such representations is an important step including considerations of observability of the known and the unknown. While a prescient approach has its limitations, like uncertainties in observations and dependence on representations, a prescient approach offers an approach

contributing to knowledge about partly unknown futures that retrospective approaches struggle to do. Expectedly, such an approach could improve the odds that management research is able to bring societal value [32] and has the possibility to revive the management discipline.

The two conjectures used as empirical examples in this paper are relating to the emergence of new technical possibilities with AI solutions in complex intelligent systems. They outline two important overall themes including a diversity of aspects that need to be addressed by management defining conditions for development and operation. The findings indicate that a prescient approach can help to find future management approaches that not only support the development of future technologies and systems but also help shaping a fruitful future with intelligent systems.

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